8 - (MathAA-S 2021-Paper 2/1-Q10) **-** *EXPONENTIALS - LOGARITHMS, DIFFERENTIATION*

[Maximum mark: 15]

Consider the function f defined by $f(x) = 90e^{-0.5x}$ for $x \in \mathbb{R}^+$.

The graph of f and the line $y = x$ intersect at point P.

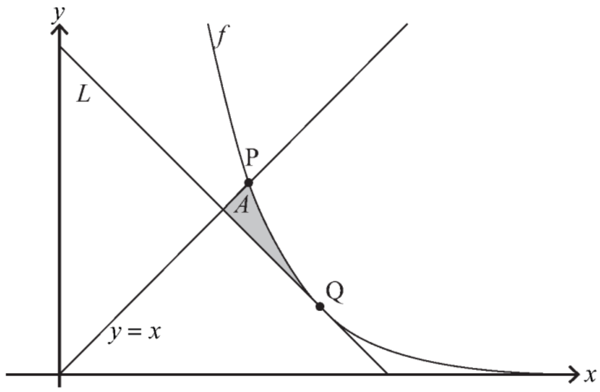
(a) Find the x-coordinate of P. [2]

The line L has a gradient of -1 and is a tangent to the graph of f at the point Q.

(b) Find the exact coordinates of Q. [4]

(c) Show that the equation of L is $y = -x + 2\ln 45 + 2$. [2]

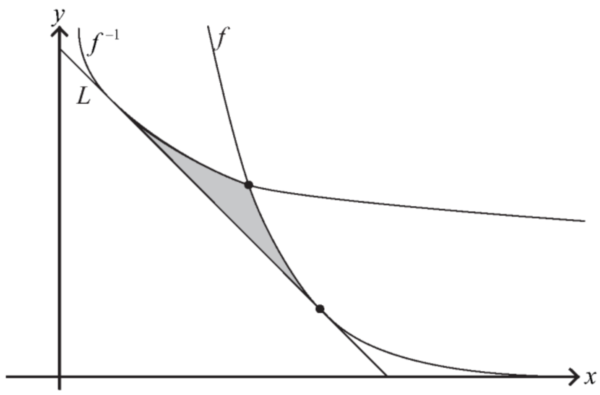
The shaded region A is enclosed by the graph of f and the lines $y = x$ and L.



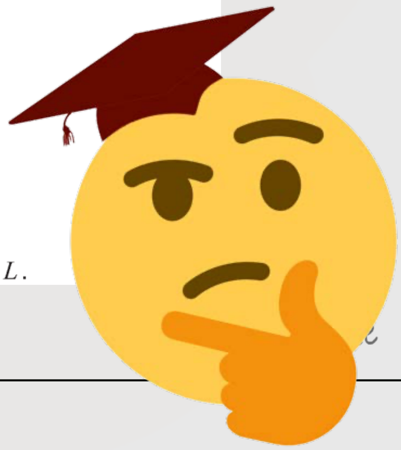
(d) (i) Find the x-coordinate of the point where L intersects the line $y = x$.

(ii) Hence, find the area of A. [5]

The line L is tangent to the graphs of both f and the inverse function f^{-1}.



(e) Find the shaded area enclosed by the graphs of f and f^{-1} and the line L.



1 - (MAT-S 2016-Paper 2/1-Q5) **-** *GRAPHS*

The function f is given by $f(x)=\dfrac{3x^2+10}{x^2-4}$, $x\in\mathbb{R}$, $x\neq 2$, $x\neq -2$.

(a) Prove that f is an even function. [2]

(b) (i) Sketch the graph $y=f(x)$.

(ii) Write down the range of f. [5]

2 - (MAT-S 2017-Paper 2/1-Q6) - *GRAPHS*

Consider the graph of $y = \tan\left(x + \frac{\pi}{4}\right)$, where $-\pi \le x \le 2\pi$.

(a) Write down the equations of the vertical asymptotes of the graph. [2]

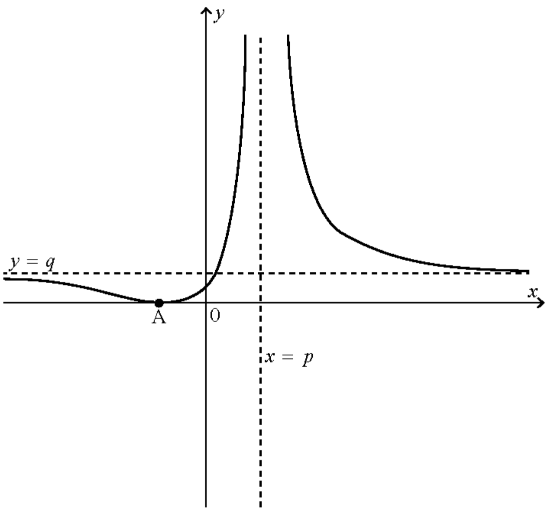
The graph is reflected in the y-axis, then stretched parallel to the y-axis by a factor $\frac{1}{2}$, then translated by $\begin{pmatrix} \frac{\pi}{4} \\ -3 \end{pmatrix}$.

(b) Give the equation of the transformed graph. [4]

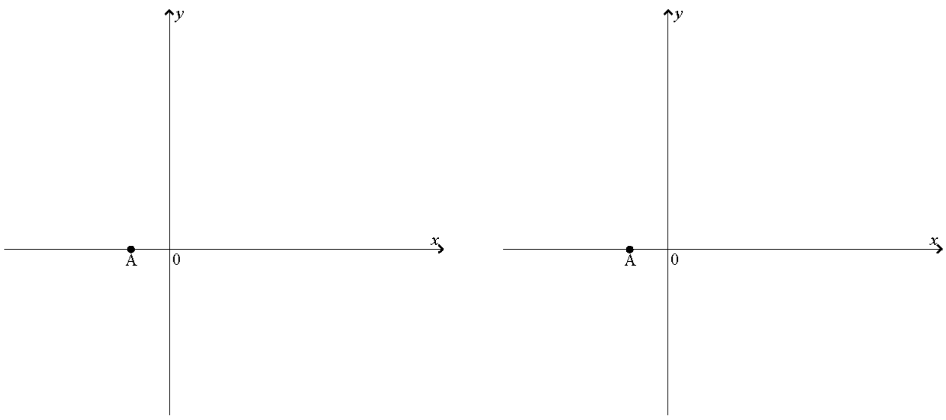
3 - (MAT-W 2018-Paper 2/0-Q8) - *GRAPHS*

Consider the function $f(x)=\dfrac{ax+1}{bx+c}$, $x \neq -\dfrac{c}{b}$, where $a, b, c \in \mathbb{Z}$.

The following graph shows the curve $y=(f(x))^2$. It has asymptotes at $x=p$ and $y=q$ and meets the x-axis at A.

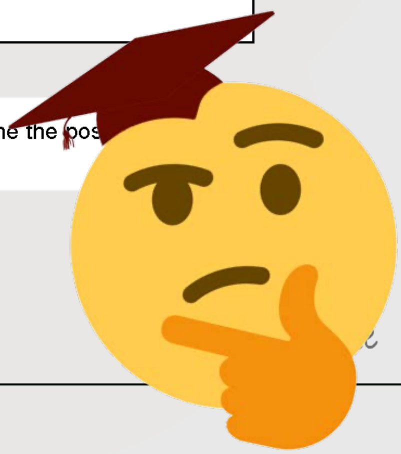


(a) On the following axes, sketch the two possible graphs of $y=f(x)$ giving the equations of any asymptotes in terms of p and q. [4]



(b) Given that $p=\dfrac{4}{3}$, $q=\dfrac{4}{9}$ and A has coordinates $\left(-\dfrac{1}{2}, 0\right)$, determine the pos

sets of values for a, b and c.



4 - (MAT-S 2019-Paper 2/2-Q9) **-** *GRAPHS*

Consider the polynomial $P(z) \equiv z^4 - 6z^3 - 2z^2 + 58z - 51$, $z \in \mathbb{C}$.

(a) Express $P(z)$ in the form $(z^2 + az + b)(z^2 + cz + d)$ where a, b, c, $d \in \mathbb{R}$. [7]

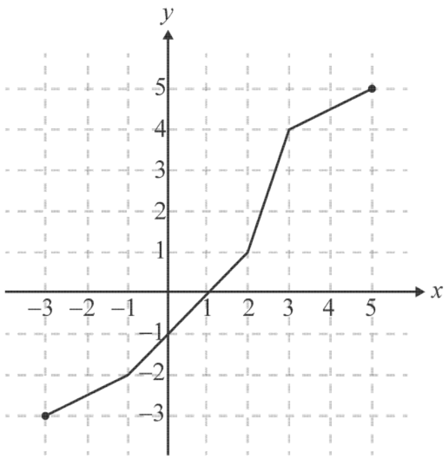
(b) Sketch the graph of $y = x^4 - 6x^3 - 2x^2 + 58x - 51$, stating clearly the coordinates of any maximum and minimum points and intersections with axes. [6]

(c) Hence, or otherwise, state the condition on $k \in \mathbb{R}$ such that all roots of the equation $P(z) = k$ are real. [2]

5 - (MAT-W 2019-Paper 2/0-Q3) **-** *GRAPHS*

[Maximum mark: 6]

The following diagram shows the graph of $y = f(x)$, $-3 \leq x \leq 5$.



(a) Find the value of $(f \circ f)(1)$. [2]

(b) Given that $f^{-1}(a) = 3$, determine the value of a. [2]

(c) Given that $g(x) = 2f(x-1)$, find the domain and range of g. [2]

6 - (MathAA-S 2021-Paper 2/1-Q11) **-** *ALGEBRA-FUNCTIONS - ROOTS, GRAPHS*

[Maximum mark: 20]

The function f is defined by $f(x) = \dfrac{3x+2}{4x^2-1}$, for $x \in \mathbb{R}$, $x \neq p$, $x \neq q$.

(a) Find the value of p and the value of q. [2]

(b) Find an expression for $f'(x)$. [3]

The graph of $y = f(x)$ has exactly one point of inflexion.

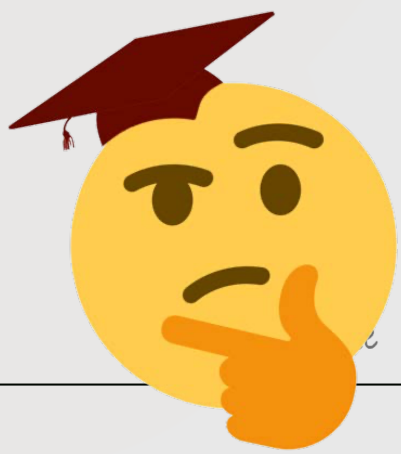
(c) Find the x-coordinate of the point of inflexion. [2]

(d) Sketch the graph of $y = f(x)$ for $-3 \leq x \leq 3$, showing the values of any axes intercepts, the coordinates of any local maxima and local minima, and giving the equations of any asymptotes. [5]

The function g is defined by $g(x) = \dfrac{4x^2-1}{3x+2}$, for $x \in \mathbb{R}$, $x \neq -\dfrac{2}{3}$.

(e) Find the equations of all the asymptotes on the graph of $y = g(x)$. [4]

(f) By considering the graph of $y = g(x) - f(x)$, or otherwise, solve $f(x) < g(x)$ for $x \in \mathbb{R}$. [4]



7 - (MathAA-W 2021-Paper 2/0-Q10) **-** *ALGEBRA-FUNCTIONS - ROOTS, GRAPHS, INTEGRATION*

[Maximum mark: 18]

Consider the function $f(x) = \dfrac{x^2 - x - 12}{2x - 15}$, $x \in \mathbb{R}$, $x \neq \dfrac{15}{2}$.

(a) Find the coordinates where the graph of f crosses the

(i) x-axis;

(ii) y-axis. [3]

(b) Write down the equation of the vertical asymptote of the graph of f. [1]

(c) The oblique asymptote of the graph of f can be written as $y = ax + b$ where $a, b \in \mathbb{Q}$.

Find the value of a and the value of b. [4]

(d) Sketch the graph of f for $-30 \leq x \leq 30$, clearly indicating the points of intersection with each axis and any asymptotes. [3]

(e) (i) Express $\dfrac{1}{f(x)}$ in partial fractions.

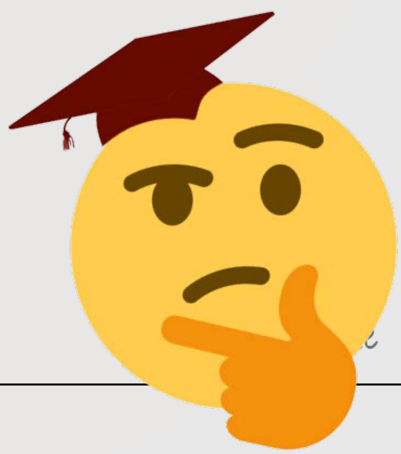
(ii) Hence find the exact value of $\displaystyle\int_0^3 \frac{1}{f(x)}\,\mathrm{d}x$, expressing your answer as a single logarithm. [7]

1 - (MAT-S 2012-Paper 2/2-Q1) **-** *SEQUENCES - SERIES*

The sum of the first 16 terms of an arithmetic sequence is 212 and the fifth term is 8.

(a) Find the first term and the common difference. *[4 marks]*

(b) Find the smallest value of n such that the sum of the first n terms is greater than 600. *[3 marks]*



2 - (MAT-S 2012-Paper 2/2-Q8) **-** *SEQUENCES - SERIES*

Each time a ball bounces, it reaches 95 % of the height reached on the previous bounce. Initially, it is dropped from a height of 4 metres.

(a) What height does the ball reach after its fourth bounce? *[2 marks]*

(b) How many times does the ball bounce before it no longer reaches a height of 1 metre? *[3 marks]*

(c) What is the total distance travelled by the ball? *[3 marks]*

3 - (MAT-W 2012-Paper 2/0-Q1) **-** *SEQUENCES - SERIES*

[Maximum mark: 4]

Find the sum of all the multiples of 3 between 100 and 500.

4 - (MAT-W 2012-Paper 2/0-Q5) **-** *SEQUENCES - SERIES*

[Maximum mark: 6]

A metal rod 1 metre long is cut into 10 pieces, the lengths of which form a geometric sequence. The length of the longest piece is 8 times the length of the shortest piece. Find, to the nearest millimetre, the length of the shortest piece.

5 - (MAT-S 2013-Paper 2/2-Q5) **-** *SEQUENCES - SERIES*

The arithmetic sequence $\{u_n : n \in \mathbb{Z}^+\}$ has first term $u_1 = 1.6$ and common difference $d = 1.5$. The geometric sequence $\{v_n : n \in \mathbb{Z}^+\}$ has first term $v_1 = 3$ and common ratio $r = 1.2$.

(a) Find an expression for $u_n - v_n$ in terms of n. *[2 marks]*

(b) Determine the set of values of n for which $u_n > v_n$. *[3 marks]*

(c) Determine the greatest value of $u_n - v_n$. Give your answer correct to four significant figures. *[1 mark]*

6 - (MAT-S 2013-Paper 2/2-Q11) **-** *SEQUENCES - SERIES, PROBABILITY*

(a) (i) Express the sum of the first n positive odd integers using sigma notation.

(ii) Show that the sum stated above is n^2.

(iii) Deduce the value of the difference between the sum of the first 47 positive odd integers and the sum of the first 14 positive odd integers. *[4 marks]*

(b) A number of distinct points are marked on the circumference of a circle, forming a polygon. Diagonals are drawn by joining all pairs of non-adjacent points.

(i) Show on a diagram all diagonals if there are 5 points.

(ii) Show that the number of diagonals is $\dfrac{n(n-3)}{2}$ if there are n points, where $n > 2$.

(iii) Given that there are more than one million diagonals, determine the least number of points for which this is possible. *[7 marks]*

(c) The random variable $X \sim B(n, p)$ has mean 4 and variance 3.

(i) Determine n and p.

(ii) Find the probability that in a single experiment the outcome is 1 or 3. *[8 marks]*

7 - (MAT-W 2013-Paper 2/0-Q2) **-** *SEQUENCES - SERIES*

The fourth term in an arithmetic sequence is 34 and the tenth term is 76.

(a) Find the first term and the common difference. *[3]*

(b) The sum of the first n terms exceeds 5000. Find the least possible value of n. *[4]*

8 - (MAT-S 2014-Paper 2/2-Q1) **-** *SEQUENCES - SERIES*

(a) (i) Find the sum of all integers, between 10 and 200, which are divisible by 7.

(ii) Express the above sum using sigma notation. *[4]*

An arithmetic sequence has first term 1000 and common difference of -6. The sum of the first n terms of this sequence is negative.

(b) Find the least value of n. *[2]*

9 - (MAT-W 2014-Paper 2/0-Q7) **-** *SEQUENCES - SERIES*

The seventh, third and first terms of an arithmetic sequence form the first three terms of a geometric sequence.

The arithmetic sequence has first term a and non-zero common difference d.

(a) Show that $d = \frac{a}{2}$. *[3]*

The seventh term of the arithmetic sequence is 3. The sum of the first n terms in the arithmetic sequence exceeds the sum of the first n terms in the geometric sequence by at least 200.

(b) Find the least value of n for which this occurs. *[6]*

10 - (MAT-S 2016-Paper 2/2-Q4) **-** *SEQUENCES - SERIES*

[Maximum mark: 6]

The sum of the second and third terms of a geometric sequence is 96.

The sum to infinity of this sequence is 500.

Find the possible values for the common ratio, r.

11 - (MAT-W 2016-Paper 2/0-Q12) - *SEQUENCES - SERIES*

On the day of her birth, 1st January 1998, Mary's grandparents invested $\$x$ in a savings account. They continued to deposit $\$x$ on the first day of each month thereafter.
The account paid a fixed rate of 0.4% interest per month. The interest was calculated on the last day of each month and added to the account.
Let $\$A_n$ be the amount in Mary's account on the last day of the nth month, immediately after the interest had been added.

(a) Find an expression for A_1 and show that $A_2 = 1.004^2x + 1.004x$. [2]

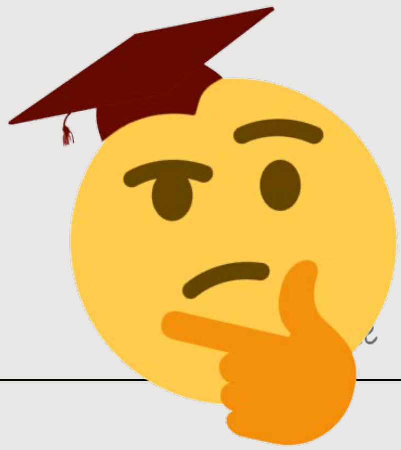
(b) (i) Write down a similar expression for A_3 and A_4.

(ii) Hence show that the amount in Mary's account the day before she turned 10 years old is given by $251(1.004^{120} - 1)x$. [6]

(c) Write down an expression for A_n in terms of x on the day before Mary turned 18 years old showing clearly the value of n. [1]

(d) Mary's grandparents wished for the amount in her account to be at least $\$20\,000$ the day before she was 18. Determine the minimum value of the monthly deposit $\$x$ required to achieve this. Give your answer correct to the nearest dollar. [4]

(e) As soon as Mary was 18 she decided to invest $\$15\,000$ of this money in an account of the same type earning 0.4% interest per month. She withdraws $\$1000$ every year on her birthday to buy herself a present. Determine how long it will take until there is no money in the account. [5]



12 - (MAT-W 2017-Paper 2/0-Q12) **-** *SEQUENCES - SERIES*

[Maximum mark: 15]

Phil takes out a bank loan of $\$150\,000$ to buy a house, at an annual interest rate of 3.5%. The interest is calculated at the end of each year and added to the amount outstanding.

(a) Find the amount Phil would owe the bank after 20 years. Give your answer to the nearest dollar. [3]

To pay off the loan, Phil makes annual deposits of $\$P$ at the end of every year in a savings account, paying an annual interest rate of 2%. He makes his first deposit at the end of the first year after taking out the loan.

(b) Show that the total value of Phil's savings after 20 years is $\dfrac{\left(1.02^{20}-1\right)P}{\left(1.02-1\right)}$. [3]

(c) Given that Phil's aim is to own the house after 20 years, find the value for P to the nearest dollar. [3]

David visits a different bank and makes a single deposit of $\$Q$, the annual interest rate being 2.8%.

(d) (i) David wishes to withdraw $\$5000$ at the end of each year for a period of n years. Show that an expression for the minimum value of Q is
$$\frac{5000}{1.028}+\frac{5000}{1.028^2}+\ldots+\frac{5000}{1.028^n}.$$

(ii) Hence or otherwise, find the minimum value of Q that would permit David to withdraw annual amounts of $\$5000$ indefinitely. Give your answer to the nearest dollar. [6]

13 - (MAT-S 2018-Paper 2/1-Q1) - *SEQUENCES - SERIES*

The 3rd term of an arithmetic sequence is 1407 and the 10th term is 1183.

(a) Find the first term and the common difference of the sequence. [4]

(b) Calculate the number of positive terms in the sequence. [3]

14 - (MAT-S 2018-Paper 2/1-Q7) **-** *SEQUENCES - SERIES*

It is known that the number of fish in a given lake will decrease by 7% each year unless some new fish are added. At the end of each year, 250 new fish are added to the lake. At the start of 2018, there are 2500 fish in the lake.

(a) Show that there will be approximately 2645 fish in the lake at the start of 2020. [3]

(b) Find the approximate number of fish in the lake at the start of 2042. [5]

15 - (MAT-W 2018-Paper 2/0-Q1) **-** *SEQUENCES - SERIES*

Consider a geometric sequence with a first term of 4 and a fourth term of -2.916.

(a) Find the common ratio of this sequence. [3]

(b) Find the sum to infinity of this sequence. [2]

16 - (MAT-S 2019-Paper 2/2-Q7) **-** *SEQUENCES - SERIES*

[Maximum mark: 7]

Suppose that u_1 is the first term of a geometric series with common ratio r.
Prove, by mathematical induction, that the sum of the first n terms, S_n is given by
$S_n = \dfrac{u_1\left(1-r^n\right)}{1-r}$, where $n \in \mathbb{Z}^+$.

17 - (MAT-W 2019-Paper 2/0-Q1) **-** *SEQUENCES - SERIES*

[Maximum mark: 5]

A geometric sequence has $u_4 = -70$ and $u_7 = 8.75$. Find the second term of the sequence.

18 - (MAT-W 2020-Paper 2/0-Q11) **-** *SEQUENCES - SERIES, KINEMATICS*

A particle P moves in a straight line such that after time t seconds, its velocity, v in $\mathrm{m\,s^{-1}}$, is given by $v = e^{-3t}\sin 6t$, where $0 < t < \frac{\pi}{2}$.

(a) Find the times when P comes to instantaneous rest. [2]

At time t, P has displacement $s(t)$; at time $t = 0$, $s(0) = 0$.

(b) Find an expression for s in terms of t. [7]

(c) Find the maximum displacement of P, in metres, from its initial position. [2]

(d) Find the total distance travelled by P in the first 1.5 seconds of its motion. [2]

At successive times when the acceleration of P is $0\,\mathrm{m\,s^{-2}}$, the velocities of P form a geometric sequence. The acceleration of P is zero at times t_1, t_2, t_3 where $t_1 < t_2 < t_3$ and the respective velocities are v_1, v_2, v_3.

(e) (i) Show that, at these times, $\tan 6t = 2$.

(ii) Hence show that $\frac{v_2}{v_1} = \frac{v_3}{v_2} = -e^{-\frac{\pi}{2}}$. [7]

19 - (MathAA-S 2021-Paper 2/2-Q2) **-** *SEQUENCES - SERIES*

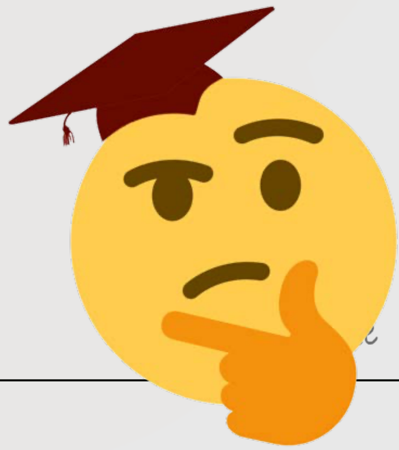
[Maximum mark: 5]

An arithmetic sequence has first term 60 and common difference -2.5.

(a) Given that the kth term of the sequence is zero, find the value of k. [2]

Let S_n denote the sum of the first n terms of the sequence.

(b) Find the maximum value of S_n. [3]



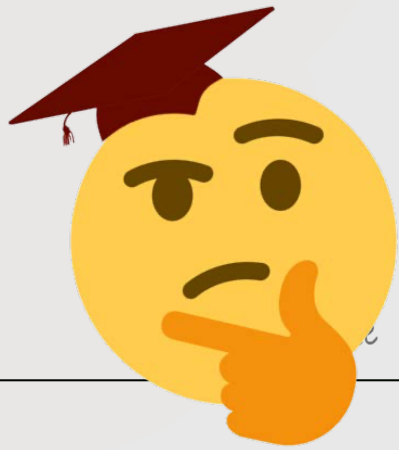
20 - (MathAA-S 2021-Paper 2/2-Q9) **-** *SEQUENCES - SERIES*

[Maximum mark: 8]

(a) Write down the first three terms of the binomial expansion of $(1+t)^{-1}$ in ascending powers of t. [1]

(b) By using the Maclaurin series for $\cos x$ and the result from part (a), show that the Maclaurin series for $\sec x$ up to and including the term in x^4 is $1+\frac{x^2}{2}+\frac{5x^4}{24}$. [4]

(c) By using the Maclaurin series for $\arctan x$ and the result from part (b), find $\lim_{x\to 0}\left(\frac{x\arctan 2x}{\sec x - 1}\right)$. [3]



21 - (MathAA-W 2021-Paper 2/0-Q5) **-** *SEQUENCES - SERIES*

[Maximum mark: 9]

The sum of the first n terms of a geometric sequence is given by $S_n = \sum_{r=1}^{n} \frac{2}{3}\left(\frac{7}{8}\right)^r$.

(a) Find the first term of the sequence, u_1. [2]

(b) Find S_∞. [3]

(c) Find the least value of n such that $S_\infty - S_n < 0.001$. [4]